

# Vectors

A **vector**,  $\mathbf{v}$ , is a mathematical object that has both **magnitude/length** and **direction**.

## Notation:

- Vectors will be written either in bold or with an arrow:  $\mathbf{v}$ ,  $\vec{v}$ .
- A vector,  $\vec{v}$ , in  $\mathbf{R}^2$  ( $\vec{v} \in \mathbf{R}^2$ ) can be represented by **2** coordinates:  
 $\vec{v} = [v_1, v_2] \in \mathbf{R}^2$
- A vector,  $\vec{v}$ , in  $\mathbf{R}^3$  ( $\vec{v} \in \mathbf{R}^3$ ) can be represented by **3** coordinates:  
 $\vec{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$
- The special unit vectors  $\mathbf{i} = [1, 0, 0]$ ,  $\mathbf{j} = [0, 1, 0]$ , and  $\mathbf{k} = [0, 0, 1]$  form a basis for  $\mathbf{R}^3$ ,  
so we can write  $\vec{v} = [v_1, v_2, v_3]$  as  $\vec{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$  (linear combination of  $\mathbf{i}$ ,  $\mathbf{j}$ ,  $\mathbf{k}$ )

## Ex:

- $[10, 14], [0, 1], [-1, 2] \in \mathbf{R}^2$  and  $[10, 4] = 10\mathbf{i} + 4\mathbf{j}$
- $[1, 1, -1], [0, 0, 0], [-10, 12, -8] \in \mathbf{R}^3$  and

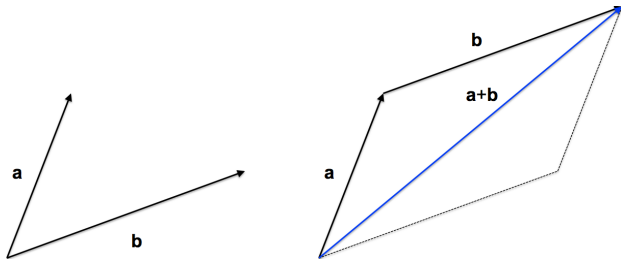
## II. Vector Addition

### Vector Operation 1: Addition

Let  $\mathbf{a} = [a_1, a_2, a_3]$  and  $\mathbf{b} = [b_1, b_2, b_3]$

Then

$$\mathbf{a} + \mathbf{b} = [a_1, a_2, a_3] + [b_1, b_2, b_3] = [a_1 + b_1, a_2 + b_2, a_3 + b_3]$$



Note:

- $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3 \implies \mathbf{a} + \mathbf{b} \in \mathbb{R}^3$
- $\mathbf{a} + \mathbf{b}$  is known as the **resultant** of  $\mathbf{a}$  and  $\mathbf{b}$ . (e.g. If  $\mathbf{a}$  and  $\mathbf{b}$  were forces then  $\mathbf{a} + \mathbf{b}$  would be the **resultant force**.)

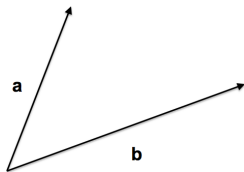
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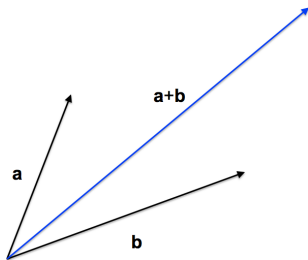
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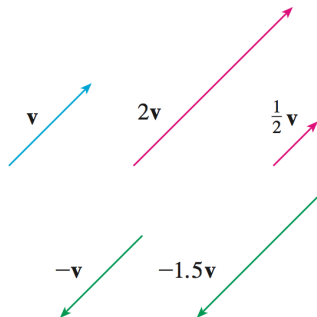
### III. Scalar Multiplication

Property: Scalar Multiplication of Vectors

Let  $\mathbf{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$  and  $c \in \mathbf{R}$  (scalar)

Then

$$c \mathbf{v} = c[v_1, v_2, v_3] = [cv_1, cv_2, cv_3]$$



Note:

- Scalar multiplication can **stretch**, **compress**, or **flip** a vector.

## IV. Length(Norm) of a Vector

Defn: Let  $\mathbf{v} = [v_1, v_2]$  and  $\mathbf{w} = [w_1, w_2, w_3]$ . Then the length of  $\mathbf{v}$  and  $\mathbf{w}$  (denoted  $|\mathbf{v}|$  and  $|\mathbf{w}|$ ) is given by

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2} \quad \text{and} \quad |\mathbf{w}| = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

Ex: Find the length of  $\mathbf{w} = [1, -2, 4]$

$$|\mathbf{w}| = \sqrt{(1)^2 + (-2)^2 + (4)^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

Note:

◦  $\mathbf{w} \in \mathbf{R}^3$  (vector), but  $|\mathbf{w}| \in \mathbf{R}$  (scalar)

Defn:  $\mathbf{v}$  is a unit vector if  $|\mathbf{v}| = 1$

Ex: Find a unit vector,  $\mathbf{u}$ , in the same direction as  $\mathbf{w}$  above.

$$\mathbf{u} = \frac{\mathbf{w}}{|\mathbf{w}|} = \frac{1}{\sqrt{21}} [1, -2, 4] = \left[ \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}, \frac{4}{\sqrt{21}} \right]$$

# Introduction

## I. Functions

- ▶ Functions have
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- ▶ Outputs live in the
- ▶ For each Input there can be
- ▶ 2 different inputs can correspond to
- ▶ 1 input can

$$F(\text{Domain}) \longrightarrow \text{Range}$$

$$F(\text{Inputs}) \longrightarrow \text{Outputs}$$

In Calculus 1 and 2, we had  $f(x)$  ...

- ▶ Domain/Input:  $x \in \mathbf{R}$  (1D line, real numbers)
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In Calculus 3, we had  $f(x, y)$  and  $f(x, y, z)$  ...

- ▶ Domain/Input:  $(x, y) \in \mathbf{R}^2$  (2D plane) or  $(x, y, z) \in \mathbf{R}^3$  (3D space)
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In This Class ... we consider  $\mathbf{F}(x)$ ,  $\mathbf{F}(x, y)$ , and  $\mathbf{F}(x, y, z)$  ...

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# Introduction

## I. Functions

- ▶ Functions have **inputs and outputs**.
- ▶ Inputs live in the **Domain**.
- ▶ Outputs live in the **Range**.
- ▶ For each Input there can be **ONLY ONE Output**.
- ▶ 2 different inputs can correspond to **1 Output**.
- ▶ 1 input can **NEVER** correspond to **2 different Outputs**.

$$F(\text{Domain}) \longrightarrow \text{Range}$$

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- $[10, 14], [0, 1], [-1, 2] \in \mathbf{R}^2$  and  $[10, 4] = 10\mathbf{i} + 4\mathbf{j}$
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Thm: Let  $\mathbf{a} = [a_1, a_2, a_3]$  and  $\mathbf{b} = [b_1, b_2, b_3]$  be vectors. Then

$$\mathbf{a} = \mathbf{b} \text{ iff}$$

Ex:  $[2, 0, 2d + c] = [c, 0, 1] \implies$

Defn: A **vector**,  $\mathbf{v}$ , is a mathematical object that has both **magnitude/length** and **direction**.

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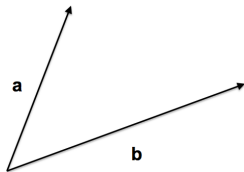
## II. Vectors & Vector Fields

### Vector Operation 1: Addition

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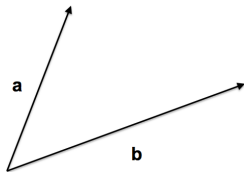
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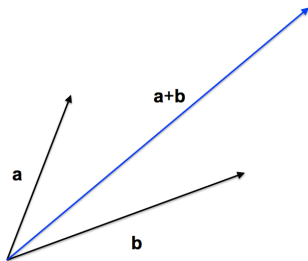
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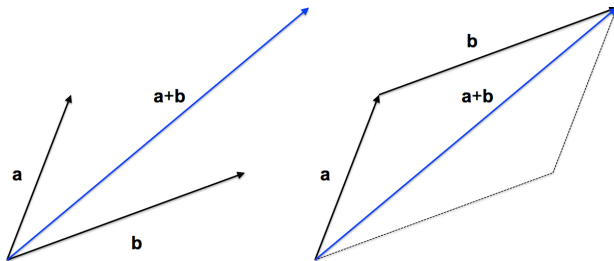
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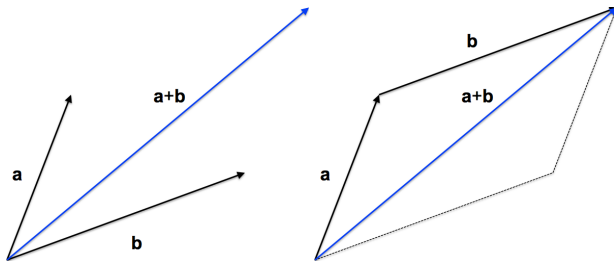
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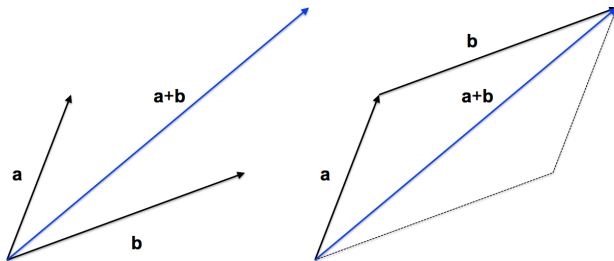
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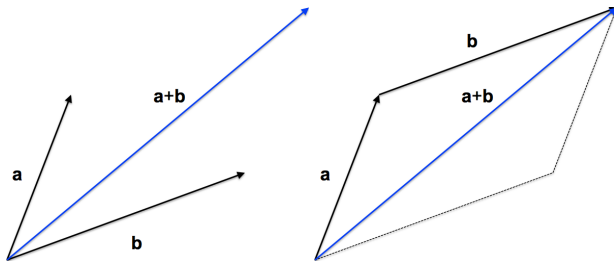
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### III. Integrals

# Vectors

## I. Vector Basics

Defn: A **vector**,  $\mathbf{v}$ , is a mathematical object that has both **magnitude/length** and **direction**.

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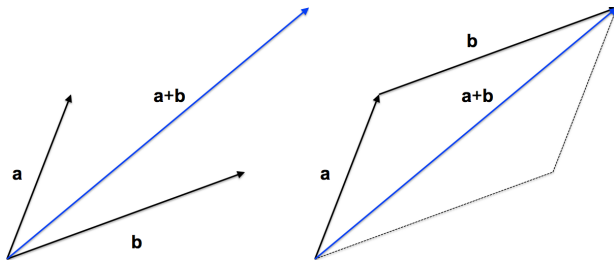
## II. Vector Addition

### Vector Operation 1: Addition

Let  $\mathbf{a} = [a_1, a_2, a_3]$  and  $\mathbf{b} = [b_1, b_2, b_3]$

Then

$$\mathbf{a} + \mathbf{b} = [a_1, a_2, a_3] + [b_1, b_2, b_3] = [a_1 + b_1, a_2 + b_2, a_3 + b_3]$$



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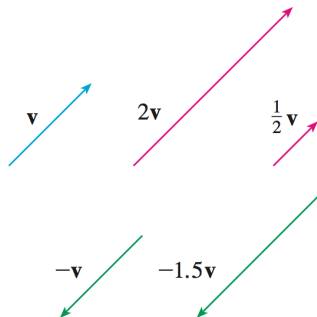
### III. Scalar Multiplication

Property: Scalar Multiplication of Vectors

Let  $\mathbf{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$  and  $c \in \mathbf{R}$  (scalar)

Then

$$c \mathbf{v} = c[v_1, v_2, v_3] = [cv_1, cv_2, cv_3]$$



Note:

- Scalar multiplication can **stretch** , **compress** , or **flip** a vector.

## IV. Length(Norm) of a Vector

Defn: Let  $\mathbf{v} = [v_1, v_2]$  and  $\mathbf{w} = [w_1, w_2, w_3]$ . Then the length of  $\mathbf{v}$  and  $\mathbf{w}$  (denoted  $|\mathbf{v}|$  and  $|\mathbf{w}|$ ) is given by

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2} \quad \text{and} \quad |\mathbf{w}| = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

Ex: Find the length of  $\mathbf{w} = [1, -2, 4]$

$$|\mathbf{w}| = \sqrt{(1)^2 + (-2)^2 + (4)^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

Note:

◦  $\mathbf{w} \in \mathbf{R}^3$  (vector), but  $|\mathbf{w}| \in \mathbf{R}$  (scalar)

Defn:  $\mathbf{v}$  is a unit vector if  $|\mathbf{v}| = 1$

Ex: Find a unit vector,  $\mathbf{u}$ , in the same direction as  $\mathbf{w}$  above.

$$\mathbf{u} = \frac{\mathbf{w}}{|\mathbf{w}|} = \frac{1}{\sqrt{21}} [1, -2, 4] = \left[ \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}, \frac{4}{\sqrt{21}} \right]$$

# Dot Product

## I. Definition

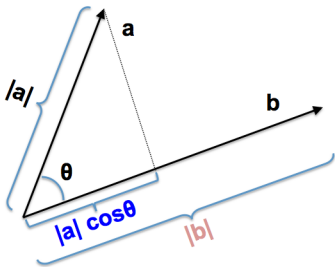
Vector Operation 2: Dot Product. Let  $\mathbf{a} = [a_1, a_2, a_3]$  and  $\mathbf{b} = [b_1, b_2, b_3]$ . Then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

- An easier way to compute the dot product of 2 known vectors is to use:

$$\mathbf{a} \cdot \mathbf{b} = [a_1, a_2, a_3] \cdot [b_1, b_2, b_3] = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Geometrical Interpretation:



Dot Product

$$\mathbf{a} \cdot \mathbf{b} = (\text{component of } \mathbf{a} \text{ along } \mathbf{b}) (\text{length of } \mathbf{b})$$

## II. Additional Concepts

Notes:

- $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3 \implies \mathbf{a} \cdot \mathbf{b} \in \mathbb{R}$  (*scalar*)

Properties: Let  $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$  be non-zero.

- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$  or equivalently  $|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$
- Commutative:  $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- Distributive:  $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$

Thm: Let  $\mathbf{a}$  and  $\mathbf{b}$  be non-zero vectors. Then

$$\mathbf{a} \cdot \mathbf{b} = 0 \quad \text{iff} \quad \mathbf{a} \text{ and } \mathbf{b} \text{ are perpendicular (orthogonal)}$$

Important Inequalities:

- Triangle Inequality:  $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$
- Cauchy-Schwarz Inequality:  $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$

### III. Application

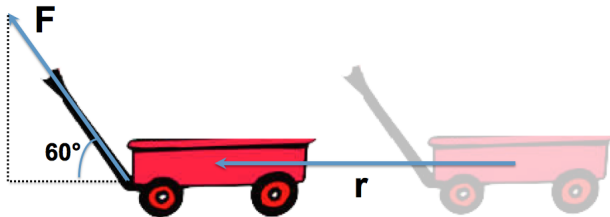
Work: In the most basic form (1-Dimension), work on an object is given by:

$$\text{Work} = \underset{\text{(on an object)}}{\text{FORCE}} \cdot \underset{\text{(object moved)}}{\text{DISTANCE}}$$

However, this assumes:

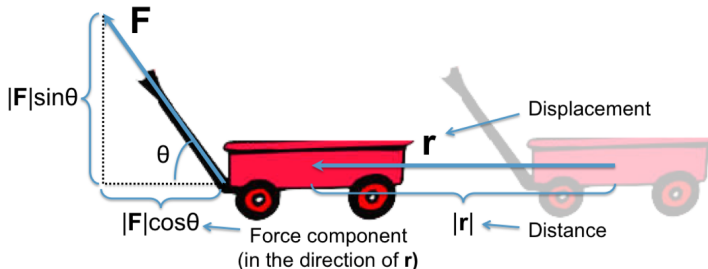
- Constant Force
- Force is acting in the same direction as the movement of the object.

Now let  $\mathbf{F}$  be a force pulling a wagon and  $\mathbf{r}$  the position of the wagon.



In this case, the force is not acting in the same direction as the movement of the wagon.

To find the work, we must decompose the force to find out how much force is contributing to the movement.



Since  $|\mathbf{F}|\cos\theta$  is the quantity of the force in the direction of movement and  $|\mathbf{r}|$  is the total distance traveled then the work to pull the wagon is

$$\text{Work} = (\text{Force})(\text{Distance}) = (|\mathbf{F}|\cos\theta)(|\mathbf{r}|) = |\mathbf{F}||\mathbf{r}|\cos\theta = \mathbf{F} \cdot \mathbf{r}$$

Therefore, for any constant force,  $\mathbf{F}$ , the general expression for work done is a **dot product**

$$\text{Work} = \mathbf{F} \cdot \mathbf{r}$$

Ex: If the wagon is being pulled by a constant force of  $[-5, 10] \text{ N}$  and the displacement of the wagon is  $[-4, 0] \text{ m}$ , then what is the work being done on the wagon?

$$\text{Work} = [-5, 10] \cdot [-4, 0] = (-5)(-4) + (10)(0) = 20 \text{ N}\cdot\text{m}$$

# Cross Product

## I. Definition

### Vector Operation 3: Cross Product

Let  $\mathbf{a} = [a_1, a_2, a_3]$ ,  $\mathbf{b} = [b_1, b_2, b_3]$ , and  $\mathbf{n}$  is a unit vector perpendicular to both  $\mathbf{a}$  and  $\mathbf{b}$

Then

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}| |\mathbf{b}| \sin \theta) \mathbf{n}$$

and so

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$$

### Notes:

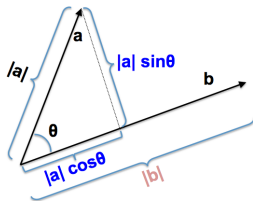
- $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3 \implies \mathbf{a} \times \mathbf{b} \in \mathbb{R}^3$  (vector) and  $\mathbf{a}, \mathbf{b} \perp \mathbf{a} \times \mathbf{b}$
- $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$  (not commutative)
- A way to remember how to calculate a vector product is to use determinants:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= (a_2 b_3 - a_3 b_2) \mathbf{i} - (a_1 b_3 - a_3 b_1) \mathbf{j} + (a_1 b_2 - a_2 b_1) \mathbf{k} \\ &= [a_2 b_3 - a_3 b_2, -a_1 b_3 + a_3 b_1, a_1 b_2 - a_2 b_1] \end{aligned}$$



## II. Additional Concepts

### Geometric Interpretation 1:



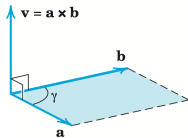
Dot Product:

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= (\text{component of } \mathbf{a} \text{ along } \mathbf{b}) (\text{length of } \mathbf{b}) \\ &= (|\mathbf{a}| \cos \theta) (|\mathbf{b}|) = |\mathbf{a}| |\mathbf{b}| \cos \theta\end{aligned}$$

Cross Product:

$$\begin{aligned}|\mathbf{a} \times \mathbf{b}| &= (\text{component of } \mathbf{a} \text{ perpendicular to } \mathbf{b}) (\text{length of } \mathbf{b}) \\ &= (|\mathbf{a}| \sin \theta) (|\mathbf{b}|) = |\mathbf{a}| |\mathbf{b}| \sin \theta\end{aligned}$$

### Geometric Interpretation 2:



$$|\mathbf{a} \times \mathbf{b}| = \text{Area of the parallelogram between } \mathbf{a} \text{ and } \mathbf{b}$$

Properties: Let  $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$  and  $d \in \mathbb{R}$  (a scalar)

- $d(\mathbf{a} \times \mathbf{b}) = d\mathbf{a} \times \mathbf{b} = \mathbf{a} \times d\mathbf{b}$
- $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
- $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$

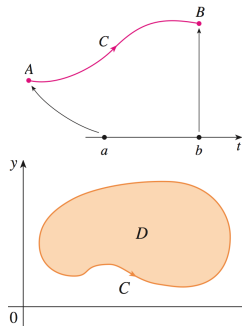
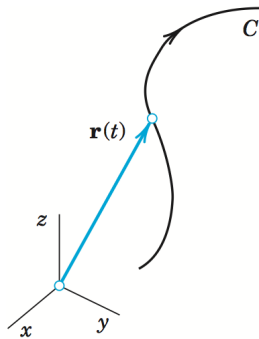
# Parametric Curves & Integration Review

## I. Parametric Curves

Defn: A **parametric curve** is a curve in space that depends on a single variable which we called the parameter. In the figure below,  $t$  is the parameter.

$$\mathbf{r}(t) = [x(t), y(t), z(t)] = x(t) \mathbf{i} + y(t) \mathbf{j} + z(t) \mathbf{k}$$

- If  $t$  is defined on a finite interval  $[a, b]$  then  $C$  will be a finite curve.
- If  $t$  is defined on a infinite interval, say  $[0, \infty]$ , then  $C$  will be a infinite or closed curve.



finite curve

closed curve

Find  $\mathbf{r}(t)$  by parametrizing the following curves

Ex:  $x = y^2 + y - 15, z = 0$  (Intersection of a parabolic cylinder and plane )

The easy parameterization is to let  $t = y$ :

$$x = t^2 + t - 15$$

$$y = t \quad \implies \quad \mathbf{r}(t) = [t^2 + t - 15, t, 0] \quad t \in \mathbf{R}$$

$$z = 0$$

Ex:  $4x - y = -1, 5x - z = 0$  (Intersection of 2 planes )

Let  $t = x$  and solve  $y$  &  $z$  in terms of  $x$  :

$$x = t$$

$$y = 4t + 1 \quad \implies \quad \mathbf{r}(t) = [t, 4t + 1, 5t] \quad t \in \mathbf{R}$$

$$z = 5t$$

Ex:  $x^2 + y^2 = 2, z = -2$  (Intersection of a cylinder and plane - circle of radius  $\sqrt{2}$   $z = -2$  )

Let  $t$  be the polar angle in the  $xy$ -plane:

$$x = \sqrt{2} \cos t$$

$$y = \sqrt{2} \sin t \quad \implies \quad \mathbf{r}(t) = [\sqrt{2} \cos t, \sqrt{2} \sin t, -2] \quad t \in [0, 2\pi]$$

Ex:  $x = y^3 + y, z = x - y$  (Intersection of a cubic cylinder and plane )

The easy parameterization is to let  $t = y$ :

$$x = t^3 + t$$

$$y = t \quad \implies \quad \mathbf{r}(t) = [t^3 + t, t, t^3] \quad t \in \mathbf{R}$$

$$z = t^3 + t - t = t^3$$

Ex:  $x^2 + y^2 = 16, z = 2x + y$  (Intersection of a cylinder and plane )

Let  $t$  be the polar angle in the  $xy$ -plane:

$$x = 4 \cos t$$

$$y = 4 \sin t \quad \implies \quad \mathbf{r}(t) = [4 \cos t, 4 \sin t, 8 \cos t + 4 \sin t] \quad t \in [0, 2\pi]$$

$$z = 8 \cos t + 4 \sin t$$

Ex: Find the velocity function given the position function  $\mathbf{r}(t) = [\underbrace{1+t^3}_{x(t)}, \underbrace{te^{-t}}_{y(t)}, \underbrace{\sin 2t}_{z(t)}]$

Recall: *velocity* =  $\mathbf{r}'(t) = [x'(t), y'(t), z'(t)]$

$$x'(t) = 3t^2$$

$$y'(t) = e^{-t} - te^{-t} \implies \mathbf{r}'(t) = [3t^2, e^{-t} - te^{-t}, 2\cos(2t)]$$

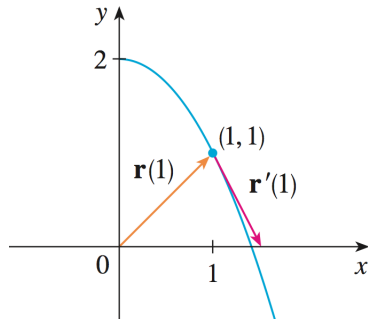
$$z'(t) = 2\cos(2t)$$

Ex: Let  $\mathbf{r}(t) = [\sqrt{t}, 2-t]$ . Find  $\mathbf{r}(1)$  and  $\mathbf{r}'(1)$

$$\mathbf{r}(1) = [\sqrt{1}, 2-1] = [1, 1]$$

$$\mathbf{r}'(t) = \left[ \frac{1}{2}t^{-1/2}, -1 \right]$$

$$\mathbf{r}'(1) = \left[ \frac{1}{2}(1)^{-1/2}, -1 \right] = \left[ \frac{1}{2}, -1 \right]$$



**Parameterizing a straight line curves.** This is simple if you know:

- (1) a vector,  $\mathbf{v}$ , parallel to the line and
- (2) a point,  $P$ , on the line.

Then the parameterization is given by:

$$\mathbf{r}(t) = \mathbf{v}t + P \quad \text{where} \quad t \in \mathbf{R}$$

Ex: Suppose we know that a line is parallel to  $[1, -1, 0]$  and passes through  $(0, 5, 5)$ . Then the parameterization for this line is:

$$\mathbf{r}(t) = [1, -1, 0]t + (0, 5, 5) \quad \text{or} \quad \mathbf{r}(t) = [t, -t + 5, 5] \quad t \in \mathbf{R}$$

Ex: Suppose we instead know that a line passes through the points  $(-3, 1, 9)$  and  $(-2, 2, 5)$ . Then we first find the vector parallel to the line by:

$$\mathbf{v} = (-3, 1, 9) - (-2, 2, 5) = [-1, -1, 4]$$

and so the parameterization is

$$\mathbf{r}(t) = [-1, -1, 4]t + (-3, 1, 9) \quad \text{or} \quad \mathbf{r}(t) = [-t - 3, -t + 1, 4t + 9] \quad t \in \mathbf{R}$$

**Parameterizing a straight line segments.** In this case we need just the initial and end points of the segment, say  $P$  and  $Q$ .

Then the parameterization is given by:

$$\mathbf{r}(t) = (Q - P)t + P \quad \text{where } t \in [0, 1]$$

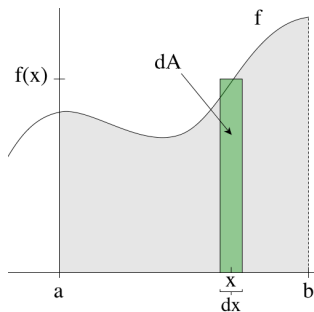
Ex: Suppose we wanted to parameterize a line with initial point  $(-1, -1, -9)$  and end point  $(10, 0, 15)$ . Then the parameterization is given by:

$$\begin{aligned} \mathbf{v} &= [(10, 0, 15) - (-1, -1, -9)] t + (-1, -1, -9) && \text{where } t \in [0, 1] \\ &= [11, 1, 24] t + (-1, -1, -9) \\ &= [11t - 1, t - 1, 24 - 9t] \end{aligned}$$

## II. Review: Integration

Remember, integration is another way of saying “add up a lot of small areas to get a whole area”. So let's think about it as such:

- ▶ Let  $dA$  be a small area (of a rectangle) and  $A$  the whole area.
- ▶ then  $dA = f(x) \cdot dx$  (see figure) and so
- ▶  $A = \int_a^b dA = \int_a^b f(x) dx$



Note: The limits of integrations ( $a$  &  $b$ ) tell you the range of the small pieces (in this example,  $dx$ )



**Interpreting Integrals:** Identify the physical meaning of the Integral

- $\iiint_E \rho(x, y, z) dV$  where  $E$  is a solid sphere and  $\rho(x, y, z)$  is the density at any point in  $E$

Total Mass of  $E$

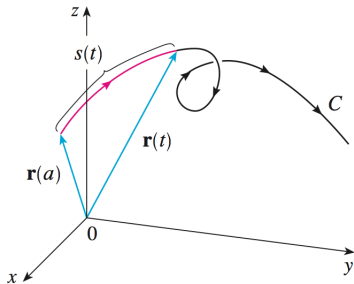
- $\int_a^b f(x) dx$  where  $f(x)$  is the force required to pull a wagon from  $a$  to  $b$

Total Work required to pull the wagon a distance  $b - a$

- $\int_C |\mathbf{v}(t)| dt$  where  $\mathbf{v}(t)$  is the velocity of a particle traveling on a finite curve  $C$

Total distance traveled by the particle along  $C$  (i.e. the arclength of  $C$ )

### III. Arclength of a Curve



Derive the integral expression for  $s(t)$ .

Assume  $C$  is parametrized by  $\mathbf{r}(t)$  for  $a \leq t \leq b$

We want to think of adding up many “small” line segments, so that we can use an integral

$$s = \int_C ds$$

But, what is  $ds$ ? ( $ds$  is a “small” distance)

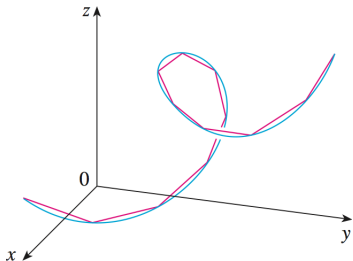
Recall: distance = (rate)(time), so

“small” distance = (rate) (“small” time)

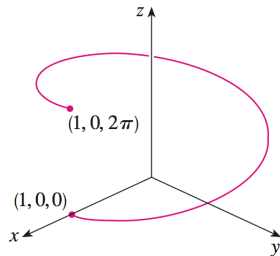
$$\text{or } ds = |\mathbf{r}'(t)| dt$$

Therefore

$$s(t) = \int_a^b |\mathbf{r}'(t)| dt$$



Compute the arclength of the curve:  $\mathbf{r}(t) = [\cos t, \sin t, t]$



Need:  $\mathbf{r}'(t)$  and  $[a, b]$  (Range of the parameter  $t$ )

Note: Since  $z = t$ , then the range of  $t$ -values is the same as the range of  $z$ -values

$$\implies t \in [0, 2\pi] \quad (\text{This is } [a, b])$$

$$\mathbf{r}'(t) = [-\sin t, \cos t, 1]$$

$$\|\mathbf{r}'(t)\|^2 = \sin^2 t + \cos^2 t + 1^2 = 1 + 1 = 2 \implies \|\mathbf{r}'(t)\| = \sqrt{2}$$

$$s(t) = \int_a^b |\mathbf{r}'(t)| dt = \int_0^{2\pi} \sqrt{2} dt = \sqrt{2} t \Big|_0^{2\pi} = 2\sqrt{2}\pi$$

# Topic: Gradient

## I. Scalar Functions

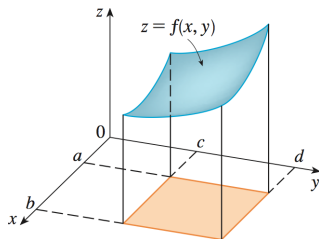
Defn: A Scalar Function is a function whose output is a scalar.

Ex:  $f(x) = x^2$  is a scalar function since  $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$

$x \in \mathbf{R}$  and  $f(x) \in \mathbf{R}$  Easy to Visualize

Ex:  $f(x, y) = x - x\sqrt{y}$  is a scalar function since  $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$

$(x, y) \in \mathbf{R}^2$  and  $f(x, y) \in \mathbf{R}$  Easy to Visualize



## II. Gradient

Defn: Let  $D_f$  be the domain of the scalar function  $f(x, y)$  and  $D_g$  be the domain of the scalar function  $g(x, y, z)$ . If  $f$  is differentiable on  $D_f$  and  $g$  is differentiable on  $D_g$  then the **gradient** of  $f(x, y)$  and  $g(x, y, z)$  is given by

$$\nabla f = \left[ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[ \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex:  $f(x, y) = xy + y$ . Find  $\nabla f$

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex:  $f(x, y, z) = z \ln(xy^2 + y)$ . Find  $\nabla f$  at the point  $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[ \frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[ \frac{(2)(1)^2}{(0)(1)^2 + (1)}, \frac{(2)(2(0)(1) + 1)}{(0)(1)^2 + (1)}, \ln((0)(1)^2 + (1)) \right] = [2, 2, 0]$$

Prove the following two properties of gradients: Recall:  $\nabla f = \left[ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1.  $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[ \frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[ \frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[ \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2.  $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= [(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z] \quad (\text{Defn of Gradient}) \\ &= [f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z] \quad (\text{Product Rule} \times 3) \\ &= [f_x \cdot g, f_y \cdot g, f_z \cdot g] + [f \cdot g_x, f \cdot g_y, f \cdot g_z] \quad (\text{Vector Addition}) \\ &= [f_x, f_y, f_z] \cdot g + f \cdot [g_x, g_y, g_z] \quad (\text{Scalar} \cdot \text{Vector}) \\ &= \nabla f \cdot g + f \cdot \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

### III. Additional Concepts

Thm:  $\nabla f(P)$  points in the direction of **maximum increase of  $f$**  at the point  $P$ .

Defn: The **Directional Derivative** of a function  $f$ ,  $D_{\mathbf{u}}f$ , in the direction of the unit vector  $\mathbf{u}$  is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

Ex: Let  $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$ ,  $P(1, 1)$  &

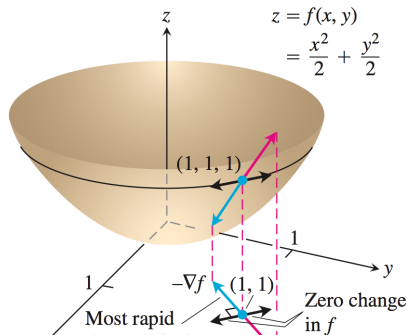
$$\mathbf{u} = \left[ \frac{1}{2}, \frac{\sqrt{3}}{2} \right]$$

Compute  $D_{\mathbf{u}}f(P)$

$\nabla f = [x, y]$  and since  $|\mathbf{u}| = 1$

$$\Rightarrow D_{\mathbf{u}}f = [x, y] \cdot \left[ \frac{1}{2}, \frac{\sqrt{3}}{2} \right] = \frac{1}{2}x + \frac{\sqrt{3}}{2}y$$

$$\Rightarrow D_{\mathbf{u}}f(P) = \frac{1}{2}(1) + \frac{\sqrt{3}}{2}(1) = \frac{1 + \sqrt{3}}{2}$$



Ex: Is there a direction  $\mathbf{u}$  in which the rate of change of the temperature function  $T(x, y) = x^2 - 3xy + 4y^2$  at  $P(1, 2)$  equals  $14^\circ\text{C}/\text{ft}$ ? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = |[-4, 13]| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$$\implies \text{Max rate of change at } P \text{ is: } \sqrt{185}$$

$$\implies \text{Min rate of change at } P \text{ is: } -\sqrt{185}$$

Therefore, since 14 does not lie within  $[\sqrt{185}, -\sqrt{185}]$ , the rate of change of the temperature can never be equal to  $14^\circ\text{C}/\text{ft}$